

# Tyler H. Chang – CV

## Research in AI, Optimization, and HPC, Founding Engineer

Mountain View, CA, USA <https://thchang.github.io>

<https://github.com/thchang> <https://scholar.google.com/citations?user=XvFLOWsAAAAJ&hl=en>

### RESEARCH INTERESTS

Numerical optimization, scientific machine learning, high-performance and parallel computing, and open source software

### EDUCATION

Ph.D., May 2020, Computer Science, Virginia Polytechnic Institute & State University (Virginia Tech)

- Advisor: Layne Watson; Dissertation: Mathematical Software for Multiobjective Optimization Problems

B.S., May 2016, Computer Science & Mathematics (double-major), Virginia Wesleyan University

- Summa cum laude; 8x Dean's list; 4x all-ODAC (conference) for varsity tennis

### RESEARCH EXPERIENCE

Jul 2026 - Present. **Founding Engineer: Stealth Mode Startup**

May 2024 - Jul 2026. **Senior Research Engineer: Siemens Digital Industry Software**, EDA / HAV Division

- Performed R&D and engineering for a high-performance EDA software
- As the senior engineer on an internal start-up-like project, I contributed to project planning, software design, implementation, and performance-tuning

Jun 2020 - May 2024. **Postdoctoral appointee: Argonne National Laboratory**, Mathematics & Computer Science Division

- Lead a portfolio of research projects related to numerical optimization and scientific machine learning
- Provided computational expertise to U.S. D.O.E. applications, including material discovery and particle accelerator design
- Designed a software framework for building and deploying parallel multiobjective simulation optimization solvers

Aug 2016 - May 2020. **Research fellow: Virginia Tech**, Dept. of Computer Science

- Researching novel methods for error-bounded scientific machine learning and numerical optimization
- Designed parallel algorithms and software for machine learning, blackbox optimization, and computational geometry
- Collaborated with scientists in HPC performance tuning, aerospace design optimization, and environmental modeling

Jun 2019 - Dec 2019. **US DOE SCGSR awardee: Argonne National Laboratory**, Mathematics & Computer Science Division

- Conducting research in multiobjective optimization software via DOE SCGSR program (see awards)

Feb 2016 - Aug 2016. **Research assistant: Old Dominion University**, Dept. of Computer Science

- Aided in parallelizing NASA's FUN3D CFD kernel on NVIDIA GPUs using CUDA and MPI

Dec 2015 - Jan 2016. **Intern: US Army Research Labs**, Computational Science Division

May 2015 - Aug 2015. **Intern: US Army Research Labs**, Computational Science Division

- Accelerating software for real-time optimal control (summer) & using AR technology for info viz (winter)

Dec 2014 - Jan 2015. **Intern: US Army Research Labs**, Guidance Technology Branch

May 2014 - Aug 2014. **Intern: US Army Research Labs**, Guidance Technology Branch

- Using OpenCV for real-time sensing (summer) & developing software for embedded systems (winter)

### PUBLICLY AVAILABLE SOFTWARE

4. 2022 (latest release 2026). ParMOO: Python library for parallel multiobjective simulation optimization. Release: 0.5.1  
Devs: **T. H. Chang** (lead), S. M. Wild, and H. Dickinson<sup>1</sup> Primary Prog. Lang: **Python 3**  
**git:** <https://github.com/parmoo/parmoo>
3. 2020 (latest release 2024). DelaunaySparse: Interpolation via a sparse subset of the Delaunay triangulation.  
Devs: **T. H. Chang** (lead), T. C. H. Lux, and L. T. Watson Primary Prog. Lang: **Fortran 2003**  
**git:** <https://github.com/vtopt/DelaunaySparse>
2. 2022. VTMOOP: Solver for blackbox multiobjective optimization problems.  
Devs: **T. H. Chang** (lead) and L. T. Watson Primary Prog. Lang: **Fortran 2008**  
**git:** <https://github.com/vtopt/VTMOOP>

<sup>1</sup>= DOE SULI (undergraduate intern) at Argonne in my supervision

1. 2019. QAML: Quantum annealing math library.  
Devs: T. C. H. Lux (lead), **T. H. Chang**, and S. S. Tipirneni      Primary Prog. Lang: **Python 3**  
**git**: <https://github.com/tchlux/qaml>



## GRANTS AND PROPOSALS

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### Completed Grants

3. Declined for FY 2024. **Key contributor** (Industry collaborator: GE Vernova, Institutional Lead: S. Chaudhuri (UIC/ANL)), \$400K/y for 1 year. *High performance computing for development of critical thermodynamic inputs for next generation thermal barrier coatings*, external grant (7 pages + appendices). DOE: HPC4EnergyInnovation Program: Summer 2024 Collaborations for U.S. Manufacturers.
2. Mar 2023 - Sep 2023. **Co-PI** (PI: G. Chen (ANL)), \$50K/y for 1 year (my share: \$25K). *A Scalable Multi-Physics Optimization Framework for Particle Accelerator Design*, institutional seed funding (3 pages + appendices). ANL LDRD: 2023 LDRD Seed (LDRD 2023-0246).
1. Jun 2019 - Dec 2019. **Primary awardee** (Advisors: J. Larson (ANL) and L. Watson (VT)), \$3K/mo for 6 months. *An Adaptive Weighting Scheme for Multiobjective Optimization*, DOE award for PhD students (3 pages + appendices). DOE SCGSR/ASCR: SCGSR Program 2018 Solicitation 2 (DE-SC0014664).

### Not Awarded

8. FY 2024. **Co-PI** (PI: Y. Li (ANL)), 2 years. *Three-dimensional antiferromagnetic memory*, institutional proposal (3 pages + appendices). ANL LDRD: 2024 LDRD Prime.
7. FY 2024. **Key personnel** (PI: C. McDevitt (Florida), Institutional Lead: J. Larson (ANL)), 4 years. *Fusion-PAD: Fusion Plant Analysis and Design*, external grant proposal (20 pages + appendices). DOE-FES/ASCR: SciDAC - FES Partnerships (DE-FOA-0002924).
6. FY 2024. **Key personnel** (PI: J. Monsalve Diaz (ANL)), 2 years. *Codelet-based Energy-aware Architectural Simulation for Future Chiplet-Enabled Supercomputing Systems*, external grant R2 pre-app (3 pages + appendices). DOE ASCR: EXPRESS - 2023 Exploratory Research for Extreme-Scale Science (DE-FOA-0002950).
5. FY 2023. **PI**, 7 months. *Assessing the Quality of Scientific Data Sets for Machine Learning*, institutional proposal (3 pages + appendices). ANL LDRD: 2023 LDRD Seed.
4. FY 2023. **Investigator** (Dir: R. Joseph (GA Tech), Institution Lead: Z. Di (ANL)), 5 years. *Statistical and Mathematical Innovations for Next Generation Facilities*, multi-institute center proposal (20 pages + appendices). DOE ASCR: 2022 Mathematical Multifaceted Integrated Capability Centers (MMICCS) (DE-FOA-0002704).
3. FY 2023. **Key personnel** (PI: G. Chin (PNNL), Institution Lead: T. Munson (ANL)), 3 years. *Visual Analytics for Complex Network Workflows to Accelerate Science into Decision Making*, external grant proposal (16 pages + appendices). DOE ASCR: Data Visualization for Scientific Discovery, Decision-Making, and Comm. (DE-FOA-0002726).
2. FY 2023. **Co-PI** (With: S. Khairy (ANL)), 8 weeks. *Accelerating Multi-Objective Neural Architecture Search*, institutional proposal (1 paragraph + admin info.). ANL CELS: 2022 LDRD Advanced Computing Expedition.
1. FY 2021. **Key personnel** (PI: A. Gillette (LLNL), Institution Lead: Z. Di (ANL)), 3 years. *FAIR Delaunay Benchmarking of Scientific AI Workflows*, external grant proposal (13 pages + appendices). DOE ASCR: FAIR Frameworks for Data and AI Models (LAB 20-2306).



## PUBLICATIONS

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### Peer-Reviewed Journal Articles

13. 2026. **T. H. Chang** and S. M. Wild. Designing a framework for solving multiobjective simulation optimization problems. *INFORMS Journal on Computing* 38(1), pp. 269–294. **doi**: 10.1287/ijoc.2023.0250
12. 2025. **T. H. Chang**, A. K. Gillette, and R. Maulik. Leveraging interpolation models and error bounds for verifiable scientific machine learning. *Journal of Computational Physics* 524, Article 113726, 23 pages. **doi**: 10.1016/j.jcp.2025.113726
11. 2024. **T. H. Chang**, L. T. Watson, S. Leyffer, T. C. H. Lux, and H. M. J. Almohri. Remark on Algorithm 1012: computing projections with large data sets. *ACM Transactions on Mathematical Software* 50(2), Article 12, 8 pages. **doi**: 10.1145/3656581
10. 2023. T. C. H. Lux, L. T. Watson, **T. H. Chang**, and W. I. Thacker. Algorithm 1031: MQSI—Monotone quintic spline interpolation. *ACM Transactions on Mathematical Software* 49(1), Article 6, 17 pages. **doi**: 10.1145/3570157
9. 2023. **T. H. Chang** and S. M. Wild. ParMOO: a Python library for parallel multiobjective simulation optimization. *Journal of Open Source Software* 8(82), Article 4468, 5 pages. **doi**: 10.21105/joss.04468
8. 2023. N. Neveu, **T. H. Chang**, P. Franz, S. Hudson, and J. Larson. Comparison of multiobjective optimization methods for the LCLS-II photoinjector. *Computer Physics Communication* 283, Article 108566, 10 pages. **doi**: 10.1016/j.cpc.2022.108566
7. 2023. Y. Wang, L. Xu, Y. Hong, R. Pan, **T. H. Chang**, T. C. H. Lux, J. Bernard, L. T. Watson, and K. W. Cameron. Design strategies and approximation methods for high-performance computing variability management. *Journal of Quality Technology* 55(1), pp. 88–103. **doi**: 10.1080/00224065.2022.2035285

6. 2022. **T. H. Chang**, L. T. Watson, J. Larson, N. Neveu, W. I. Thacker, S. Deshpande, and T. C. H. Lux. Algorithm 1028: VTMOPT: Solver for blackbox multiobjective optimization problems. *ACM Transactions on Mathematical Software* 48(3), Article 36, 34 pages. **doi**: 10.1145/3529258
5. 2021. L. Xu, T. C. H. Lux, **T. H. Chang**, B. Li, Y. Hong, L. T. Watson, A. R. Butt, D. Yao, and K. W. Cameron. Prediction of high-performance computing input/output variability and its application to optimization for system configurations. *Quality Engineering* 33(2), pp. 318–334. **doi**: 10.1080/08982112.2020.1866203
4. 2021. T. C. H. Lux, L. T. Watson, **T. H. Chang**, J. Bernard, B. Li, L. Xu, G. Back, A. R. Butt, K. W. Cameron, and Y. Hong. Interpolation of sparse high-dimensional data. *Numerical Algorithms* 88(1), pp. 281–313. **doi**: 10.1007/s11075-020-01040-2
3. 2020. **T. H. Chang**, L. T. Watson, T. C. H. Lux, A. R. Butt, K. W. Cameron, and Y. Hong. Algorithm 1012: DELAUNAYSPARSE: Interpolation via a sparse subset of the Delaunay triangulation in medium to high dimensions. *ACM Transactions on Mathematical Software* 46(4), Article 38, 20 pages. **doi**: 10.1145/3422818
2. 2020. L. Xu, Y. Wang, T. C. H. Lux, **T. H. Chang**, J. Bernard, B. Li, Y. Hong, K. W. Cameron, and L. T. Watson. Modeling I/O performance variability in high-performance computing systems using mixture distributions. *Journal of Parallel and Distributed Computing* 139, pp. 87–98. **doi**: 10.1016/j.jpdc.2020.01.005
1. 2019. **T. H. Chang**, T. C. H. Lux, and S. S. Tipirneni. Least-squares solutions to polynomial systems of equations with quantum annealing. *Quantum Information Processing* 18(12), Article 374, 17 pages. **doi**: 10.1007/s11128-019-2489-x

## Peer-Reviewed Conference and Workshop Papers

15. 2023. G. Chen, **T. H. Chang**, J. Power, and C. Jing. An integrated multi-physics optimization framework for particle accelerator design. In *Proc. 2023 Winter Simulation Conference (WSC 2023), Industrial Applications Track*, 2 pages. Orlando, FL, USA. **doi**: 10.48550/arXiv.2311.09415
14. 2023. M. Garg<sup>2</sup>, **T. H. Chang**, and K. Raghavan. SF-SFD: Stochastic optimization of Fourier coefficients for space-filling designs. In *Proc. 2023 Winter Simulation Conference (WSC 2023)*, pp. 3636–3646. Orlando, FL, USA. **doi**: 10.1109/WSC60868.2023.10408245
13. 2023. **T. H. Chang**, J. R. Elias, S. M. Wild, S. Chaudhuri, and J. A. Libera. A framework for fully autonomous design of materials via multiobjective optimization and active learning: challenges and next steps. In *11th Intl. Conf. on Learning Representation (ICLR 2023), Workshop on Machine Learning for Materials (ML4Materials)*. Kigali, Rwanda. **url**: <https://openreview.net/forum?id=8KJS7RPjMqG>
12. 2020. **T. H. Chang**, J. Larson, and L. T. Watson. Multiobjective optimization of the variability of the high-performance Linpack solver. In *Proc. 2020 Winter Simulation Conference (WSC 2020)*, pp. 3081–3092. Orlando, FL, USA. **doi**: 10.1109/WSC48552.2020.9383875
11. 2020. **T. H. Chang**, J. Larson, L. T. Watson, and T. C. H. Lux. Managing computationally expensive blackbox multiobjective optimization problems with libEnsemble. In *Proc. 2020 Spring Simulation Conference (SpringSim '20)*, Article 31, 12 pages. Fairfax, VA, USA. **doi**: 10.22360/springsim.2020.hpc.001
10. 2020. T. C. H. Lux, L. T. Watson, **T. H. Chang**, L. Xu, Y. Wang, and Y. Hong. An algorithm for constructing monotone quintic interpolating splines. In *Proc. 2020 Spring Simulation Conference (SpringSim '20)*, Article 33, 12 pages. Fairfax, VA, USA. **doi**: 10.22360/springsim.2020.hpc.003
9. 2020. T. C. H. Lux and **T. H. Chang**. Analytic test functions for generalizable evaluation of convex optimization techniques. In *Proc. IEEE SoutheastCon 2020*, 8 pages. Raleigh, NC, USA. **doi**: 10.1109/SoutheastCon44009.2020.9368254
8. 2020. T. C. H. Lux, L. T. Watson, **T. H. Chang**, L. Xu, Y. Wang, J. Bernard, Y. Hong, and K. W. Cameron. Effective nonparametric distribution modeling for distribution approximation applications. In *Proc. IEEE SoutheastCon 2020*, 6 pages. Raleigh, NC, USA. **doi**: 10.1109/SoutheastCon44009.2020.9368295
7. 2018. **T. H. Chang**, L. T. Watson, T. C. H. Lux, S. Raghvendra, B. Li, L. Xu, A. R. Butt, K. W. Cameron, and Y. Hong. Computing the umbrella neighbourhood of a vertex in the Delaunay triangulation and a single Voronoi cell in arbitrary dimension. In *Proc. IEEE SoutheastCon 2018*, 8 pages. St. Petersburg, FL, USA. **doi**: 10.1109/SECON.2018.8479003
6. 2018. T. C. H. Lux, L. T. Watson, **T. H. Chang**, J. Bernard, B. Li, X. Yu, L. Xu, G. Back, A. R. Butt, K. W. Cameron, Y. Hong, and D. Yao. Nonparametric distribution models for predicting and managing computational performance variability. In *Proc. IEEE SoutheastCon 2018*, 7 pages. St. Petersburg, FL, USA. **doi**: 10.1109/SECON.2018.8478814
5. 2018. **T. H. Chang**, L. T. Watson, T. C. H. Lux, J. Bernard, B. Li, L. Xu, G. Back, A. R. Butt, K. W. Cameron, and Y. Hong. Predicting system performance by interpolation using a high-dimensional Delaunay triangulation. In *Proc. 2018 Spring Simulation Conference (SpringSim '18)*, Article 2, 12 pages. Baltimore, MD, USA. **doi**: 10.22360/springsim.2018.hpc.003
4. 2018. T. C. H. Lux, L. T. Watson, **T. H. Chang**, J. Bernard, B. Li, L. Xu, G. Back, A. R. Butt, K. W. Cameron, and Y. Hong. Predictive modeling of I/O characteristics in high performance computing systems. In *Proc. 2018 Spring Simulation Conference (SpringSim '18)*, Article 8, 10 pages. Baltimore, MD, USA. **doi**: 10.22360/springsim.2018.hpc.009
3. 2018. **T. H. Chang**, L. T. Watson, T. C. H. Lux, B. Li, L. Xu, A. R. Butt, K. W. Cameron, and Y. Hong. A polynomial time algorithm for multivariate interpolation in arbitrary dimension via the Delaunay triangulation. In *Proc. 2018 ACM Southeast Conference (ACMSE '18)*, Article 12, 8 pages. Richmond, KY, USA. **doi**: 10.1145/3190645.3190680
2. 2018. T. C. H. Lux, L. T. Watson, **T. H. Chang**, J. Bernard, B. Li, X. Yu, L. Xu, G. Back, A. R. Butt, K. W. Cameron, D. Yao, and Y. Hong. Novel meshes for multivariate interpolation and approximation. In *Proc. 2018 ACM Southeast Conference (ACMSE '18)*, Article 13, 7 pages. Richmond, KY, USA. **doi**: 10.1145/3190645.3190687

<sup>2</sup>= NSF MSGI (PhD student intern) at Argonne in my supervision

1. 2017. C. Raghunath, **T. H. Chang**, L. T. Watson, M. Jrad, R. K. Kapania, and R. M. Kolonay. Global deterministic and stochastic optimization in a service oriented architecture. In *Proc. 2017 Spring Simulation Conference (SpringSim '17)*, Article 7, 12 pages. Virginia Beach, VA, USA. **doi** : 10.22360/springsim.2017.hpc.023

## Technical Reports and Dissertations

5. 2026. **T. H. Chang**, S. M. Wild, and H. Dickinson<sup>1</sup>. *ParMOO: Python library for parallel multiobjective simulation optimization*. Technical Report Version 0.5.1. Argonne National Laboratory, Lemont, Illinois, USA. **url** : [https://parmoo.readthedocs.io/\\_/downloads/en/latest/pdf/](https://parmoo.readthedocs.io/_/downloads/en/latest/pdf/)
4. 2023. R. Egele, **T. H. Chang**, Y. Sun, V. Vishwanath, and P. Balaprakash. *Parallel multi-objective hyperparameter optimization with uniform normalization and bounded objectives*. Technical Report arXiv:2309.14936 [cs.LG]. arXiv preprint. Submitted Apr 2024. Preprint: <https://arxiv.org/abs/2309.14936>. **url** : <https://arxiv.org/abs/2309.14936>
3. 2020. A. K. Gillette and **T. H. Chang**. *ALGORITHMS: assessing latent space dimension by Delaunay loss*. Technical Report LLNL-CONF-814930. Lawrence Livermore National Laboratory, Livermore, California, USA. **url** : <https://www.osti.gov/servlets/purl/1756748>
2. 2020. **T. H. Chang**. *Mathematical software for multiobjective optimization problems*. Ph.D. Dissertation. Department of Computer Science, Virginia Polytechnic Institute and State University (Virginia Tech), Blacksburg, VA, USA. **Dept. of Computer Science Nominee for Outstanding Dissertation Award**. **url** : <http://hdl.handle.net/10919/98915>
1. 2016. **T. H. Chang**. *GPU saturation for multiple matrix-vector multiplications*. Bachelor's Thesis. Department of Computer Science, Virginia Wesleyan University, Virginia Beach, VA, USA.



## TALKS AND PRESENTATIONS

### Invited Conference Talks

7. Jan 2026. Propagating uncertainty information through gray-box multiobjective optimization problems. *Dagstuhl Seminar 26041: Uncertainty Quantification in Multiobjective Optimization*, Wadern, Germany.
6. Aug 2023. Data sampling for surrogate modeling and optimization. *The 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023)*, Tokyo, Japan.
5. Jun 2023. Exploiting structures in multiobjective simulation optimization problems. *SIAM Conference on Optimization (OP 2023)*, Seattle, WA, USA.
4. Mar 2023. ParMOO: a Python library for parallel multiobjective simulation optimization. *SIAM Conference on Computational Science and Engineering (CSE 2023)*, Amsterdam, Netherlands.
3. Sep 2022. Geometric considerations when surrogate modeling. *SIAM Conference on Mathematics of Data Science (MDS 2022)*, San Diego, CA, USA.
2. Jul 2021. Surrogate modeling of simulations for multiobjective optimization applications. *SIAM Conference on Optimization (OP 2021)*, virtual event.
1. Mar 2021. Computing sparse subsets of the Delaunay triangulation in high-dimensions for interpolation and graph problems. *SIAM Conference on Computational Science and Engineering (CSE 2021)*, virtual event.

### Seminars and Colloquia

5. Sep 2023. The curse of dimensionality. *CAA&CM Argonne Student Visit*, Lemont, IL, USA.
4. July 2023. Toward interpretable machine learning via Delaunay interpolation – challenges and next steps. *Argonne National Laboratory, LANS Seminar Series*, Lemont, IL, USA.
3. Feb 2020. Algorithms and software for Delaunay interpolation and multiobjective optimization. *Sandia National Laboratory, Wind Energy Technology Division Seminar*, virtual event.
2. Feb 2020. Algorithms and software for Delaunay interpolation and multiobjective optimization. *Argonne National Laboratory, Mathematics and Computer Science Division Seminar*, Lemont, IL, USA.
1. Jan 2020. Algorithms and software for Delaunay interpolation and multiobjective optimization. *Sandia National Laboratory, Center for Computing Research Seminar*, Albuquerque, NM, USA.

### Tutorials and Guest Lectures

2. Oct 2022. An introduction to multiobjective simulation optimization with ParMOO. *The Science Academy, Science Circle Cohort, guest speaker*, virtual event. Recording: <https://www.youtube.com/watch?v=gqha8urlehm>.
1. May 2022. An introduction to multiobjective simulation optimization with ParMOO. *University of Chicago, Pritzker School of Molecular Engineering, guest lecture*, virtual event.

### Contributed Talks

13. Oct 2025. Leveraging interpolation models and error bounds for verifiable scientific machine learning. *2nd SIAM Northern and Central California Sectional Conference (NCC25)*, Berkeley, CA, USA.
12. Dec 2023. An integrated multi-physics optimization framework for particle accelerator design. *2023 Winter Simulation Conference (WSC 2023)*, San Antonio, TX, USA.

11. Jul 2022. ParMOO: a parallel framework for multiobjective simulation optimization problems. *Seventh International Conference on Continuous Optimization (ICCOPT 2022)*, Bethlehem, PA, USA.
10. Dec 2020. Multiobjective optimization of the variability of the high-performance Linpack solver. *2020 Winter Simulation Conference (WSC 2020)*, virtual event.
9. May 2020. Managing computationally expensive blackbox multiobjective optimization problems with libEnsemble. *Spring Simulation Conference (SpringSim '20)*, virtual event.
8. Aug 2019. A surrogate for local optimization using Delaunay triangulations. *Sixth International Conference on Continuous Optimization (ICCOPT 2019)*, Berlin, Germany.
7. Apr 2018. Computing the umbrella neighbourhood of a vertex in the Delaunay triangulation and a single Voronoi cell in arbitrary dimension. *IEEE SoutheastCon 2020*, St. Petersburg, FL, USA.
6. Apr 2018. Nonparametric distribution models for predicting and managing computational performance variability. *IEEE SoutheastCon 2020*, St. Petersburg, FL, USA.
5. Mar 2018. A polynomial time algorithm for multivariate interpolation in arbitrary dimension via the Delaunay triangulation. *ACM Southeast Conference (ACMSE '18)*, Richmond, KY, USA.
4. Apr 2017. Global deterministic and stochastic optimization in a service oriented architecture. *Spring Simulation Conference (SpringSim '17)*, Virginia Beach, VA, USA.
3. Apr 2016. GPU saturation for multiple matrix-vector multiplications. *Virginia Wesleyan College, Undergraduate Research Symposium*, Virginia Beach, VA, USA.
2. Aug 2015. Implementing the optimal control-based obstacle avoidance (OCA) algorithm in compiled code. *ARL Summer Student Research Symposium Compendium of Abstracts*, Aberdeen, MD, USA.
1. Apr 2014. The new Mastermind. *Virginia Wesleyan College, Undergraduate Research Symposium*, Virginia Beach, VA, USA.

## Posters and Lightning Talks

10. May 2023. A framework for fully autonomous design of experiments via active learning and multiobjective optimization: challenges and next steps (poster). *11th Intl. Conf. on Learning Representation (ICLR 2023), Workshop on Machine Learning for Materials (ML4Materials)*, Kigali, Rwanda.
9. Oct 2022. ParMOO: a Python library for parallel multiobjective simulation optimization (lightning talk). *FASTMath All-Hands Meeting*, Denver, CO, USA.
8. May 2022. ParMOO: parallel multiobjective optimization in python (poster). *The Argonne, Chicago, Michigan, Northwestern, Toyota, Wisconsin (ACMNTW) Workshop on Optimization and Machine Learning*, Evanston, IL, USA.
7. Nov 2020. Multiobjective optimization of simulations with PARMOO (lightning talk). *Argonne Postdoctoral Research and Career Symposium*, virtual event.
6. Feb 2020. VarSys: managing variability for high performance computers, cloud computing systems, and computer security (poster). *Virginia Tech, Graduate Student Recruitment Weekend, Lab Poster Session*, Blacksburg, VA, USA.
5. Aug 2019. An adaptive weighting scheme for multiobjective optimization (lightning talk). *Argonne National Laboratory, LANS SASSy Seminar Series*, Lemont, IL, USA.
4. Mar 2019. VarSys: managing variability for high performance computers, cloud computing systems, and computer security (poster). *Virginia Tech, Graduate Student Recruitment Weekend, Lab Poster Session*, Blacksburg, VA, USA.
3. Sep 2018. A polynomial time algorithm for multivariate interpolation in arbitrary dimension via the Delaunay triangulation (lightning talk). *Virginia Tech, CS GSC: Lightning Talks*, Blacksburg, VA, USA.
2. Mar 2018. VarSys: managing variability for high performance computers, cloud computing systems, and computer security (poster). *Virginia Tech, Graduate Student Recruitment Weekend, Lab Poster Session*, Blacksburg, VA, USA.
1. Aug 2014. Real-time target identification in Matlab with OpenCV (poster). *US Army Research Laboratories, Summer Student Poster Symposium*, Aberdeen, MD, USA.



## AWARDS AND FELLOWSHIPS

### Research Fellowships

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|----------------------|-------------------------------------------------------------------------------------------------|
| Aug 2016 - May 2020. | Cunningham Doctoral Fellowship, Virginia Tech, Graduate School, guaranteed research funding     |
| Aug 2019 - May 2020. | Davenport Leadership Fellowship, Virginia Tech, College of Engineering, \$4k supplemental award |
| Aug 2018 - May 2019. | Pratt Fellowship, Virginia Tech, College of Engineering, \$4k supplemental award                |
| Aug 2017 - May 2018. | Pratt Fellowship, Virginia Tech, College of Engineering, \$4k supplemental award                |
| Aug 2016 - May 2017. | Davenport Leadership Fellowship, Virginia Tech, College of Engineering, \$4k supplemental award |

### Awards

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|-----------|-------------------------------------------------------------------------------------------------|
| Jan 2021. | Nominee for Outstanding Dissertation Award, Virginia Tech, Graduate School                      |
| Apr 2016. | Outstanding Student in Computer Science & Mathematics, Virginia Wesleyan University             |
| Feb 2016. | ACM International Collegiate Programming Competition (ICPC), winning team for CNU site, VA, USA |
| Feb 2015. | ACM International Collegiate Programming Competition (ICPC), winning team for CNU site, VA, USA |

## Misc. Travel Awards

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|-----------|------------------------------------------------------------------------------------------------|
| Oct 2025. | SIAM Early Career Travel Award, \$400 to attend SIAM NCC 25                                    |
| Feb 2023. | SIAM Early Career Travel Award, \$1750 to attend ICIAM 2023                                    |
| Feb 2021. | SIAM Early Career Travel Award, \$160 to cover SIAM CSE 2021 registration                      |
| Apr 2020. | Virginia Tech Computer Science Graduate Travel Award, \$400 to attend SpringSim 2020           |
| Aug 2019. | Virginia Tech Graduate Student Association Travel Fund, \$225 to attend ICCOPT 2019            |
| Jun 2019. | Virginia Tech Computer Science Graduate Travel Award, \$600 to attend ICCOPT 2019              |
| Apr 2019. | ICCOPT Travel Grant Program, €450 to attend ICCOPT 2019                                        |
| Apr 2018. | Virginia Tech Graduate Student Association Travel Fund, \$225 to attend IEEE SoutheastCon 2018 |
| Mar 2018. | Virginia Tech Computer Science Graduate Travel Award, \$400 to attend IEEE SoutheastCon 2018   |
| Aug 2016. | Virginia Tech Cunningham Travel Fund, \$1000 unrestricted travel fund for Cunningham fellow    |

## TEACHING EXPERIENCE

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|----------------------|--------------------------------------------------------------------------------------------------------|
| Jan 2022 - Feb 2024. | <b>Adjunct Professor:</b> College of DuPage, Dept. of Computer and Info. Science (Intro to Python)     |
| Jan 2020 - May 2020. | <b>Instructor of Record:</b> Virginia Tech, Dept. of Computer Science (Data structures and algorithms) |
| Jan 2013 - Dec 2015. | <b>Subject Tutor:</b> Virginia Wesleyan University (Calculus, computer science, and statistics)        |

## ADVISING

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|----------------------|--------------------------------------------------------------------|
| Jun 2022 - Aug 2022. | Manisha Garg (UIUC), NSF MSGI (PhD student intern) at Argonne      |
| Jun 2022 - Aug 2022. | Hyrum Dickinson (UIUC), DOE SULI (undergraduate intern) at Argonne |

## MENTORING

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|-----------|--------------------------------------------------------------------------------|
| Aug 2022. | Srinivas Eswar (postdoc at Argonne), via Argonne Navigator Program             |
| Mar 2021. | Bosco Ndemeye (PhD student at Univ. of Oregon), via SIAM CSE Mentoring Program |

## PROFESSIONAL SERVICE AND ACTIVITIES

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### Journal Referee

Journal of Open Source Software (2025–Present); Optimization Methods and Software (2025–Present); Journal of Supercomputing (2024–Present); INFORMS Journal on Computing (2023–Present); ACM Transactions on Mathematical Software (2021–Present); Quantum Information Processing (2021–Present); The Visual Computer Journal (2021); MDPI: Mathematical and Computer Applications (2021); Journal of Machine Learning Research (2019)

### Conference Reviewer

ICIAM (2023); Supercomputing (2021); IEEE SoutheastCon (2018–2020)

### Minisymposium Organizer

Multiobjective Optimization Software track in SIAM Conference on Optimization (2021); Geometric Methods for Machine Learning track in SIAM Conference on Computational Science and Engineering (2021)

### Professional Membership

ACM (2015–Present); SIAM (2016–Present)

## INSTITUTIONAL SERVICE

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### Siemens

|                     |                                                                            |
|---------------------|----------------------------------------------------------------------------|
| Jan 2026 - Present. | Member of advisory committee for defining AI Architect Job Role at Siemens |
| May 2025 - Present. | Representative of Siemens EDA at SystemX alliance workshop at Stanford     |



## Argonne

|                      |                                                         |
|----------------------|---------------------------------------------------------|
| Nov 2022 - Present.  | Organizing Committee, FASTMath Institute Seminar Series |
| Dec 2022 - Jan 2023. | Hiring Committee, MCS Division                          |
| Oct 2022 - Dec 2022. | Technical Reviewer, Sustainable Research Pathways       |
| Mar 2022 - Apr 2022. | Red Team Reviewer, ASCR Pre-proposals                   |

## College of DuPage

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| Jan 2024 - May 2024. | Member of Computer and Information Science Advisory Board |
| Aug 2023 - Sep 2023. | Faculty contact for Argonne visit                         |

## Virginia Tech

|                      |                                                                   |
|----------------------|-------------------------------------------------------------------|
| Aug 2017 - May 2020. | Founding Member, Computer Science Graduate Counsel                |
| Dec 2018 - Mar 2019. | Lead Student Organizer, Computer Science Graduate Preview Weekend |



## POST-DISSERTATION COURSES AND CERTIFICATIONS

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|           |                                                                                                |
|-----------|------------------------------------------------------------------------------------------------|
| Jul 2022. | Distributionally Robust Optimization, ICCOPT Summer School                                     |
| Jul 2022. | Mathematical Optimization in Julia with JuMP, ICCOPT Summer School                             |
| Aug 2020. | Probability, Geometry, and Computation in High Dimensions Boot Camp, Simons Institute Workshop |



## REFERENCES

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Andrew Gillette (Computational Scientist, Lawrence Livermore National Lab)

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Santanu Chaudhuri (Professor of Civil, Material, and Environmental Engineering, University of Illinois Chicago)

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Kirk Cameron (Professor of Computer Science, Virginia Tech)

- Former project PI
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